Alex Rivera

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Target Countries for Graduate Study: Germany, Switzerland, United Kingdom, United States

EDUCATION

Bachelor of Technology in Computer Science & Engineering
Institute of Technology & Sciences (2021 - 2025)
Cumulative Grade Point Average (CGPA): 8.92 / 10.0 (Top 5% of graduating class)

TECHNICAL SKILLS

Languages & Runtimes: Python, C++, Go, TypeScript, SQL
Frameworks & Tools: FastAPI, PyTorch, LangGraph, Docker, Kubernetes, PostgreSQL
Core Competencies: Distributed Systems, Machine Learning Systems, High-Performance Computing

SELECTED RESEARCH & ENGINEERING PROJECTS

1. Edge-AI Inference Accelerator (Python, C++, TensorRT)
- Designed an open-source low-latency quantized neural inference pipeline for resource-constrained edge hardware.
- Reduced memory footprint by 42% and achieved 3.2x throughput increase on Jetson platforms.
- Published codebase with 140+ GitHub stars and complete documentation.

2. Distributed Vector Index for Semantic Retrieval (Go, gRPC, Raft)
- Engineered a fault-tolerant vector clustering service with HNSW graph indexing and Raft consensus.
- Sustained 4,500 queries per second with p99 latency under 12ms across a 4-node cluster.

ACADEMIC HONORS & ACHIEVEMENTS

- Winner, National Inter-University AI Innovation Hackathon (1st place among 120 teams, 2024)
- Dean's Merit Scholarship Award for Academic Distinction (consecutive 3 years: 2022, 2023, 2024)
- Author of peer-reviewed workshop paper on Edge Intelligence at IEEE Student Symposium 2024